Algorithms and Programming

NAVIGAZIONE QUIZ

1 2 3 4 5 6 7 8 9 10 11 12

Successivo

Tempo rimasto **1:18:38**

Avvia una nuova anteprima

NAVIGAZIONE

Dashboard
- Home del sito
- Pagine del sito
- Corso in uso
- I miei corsi

AMMINISTRAZIONE

- Gestione Quiz
- Amministrazione del corso
- Cambia ruolo in ...

Puoi visualizzare l'anteprima del quiz, tuttavia se questo fosse un tentativo reale, non ti sarà possibile in quanto:

Spiacente, il quiz non è disponibile

Domanda 1
Risposta non ancora data
Punteggio max.: 1,00
Contrassegna domanda
Modifica domanda

Given the following array of integer values, sort it in descending order using the selection sort procedure.

```
1 4 8 -11 5 3 1 7 4 3 2 12 15
```

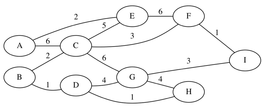
Display the content of the array just after 6 iterations of the main (outer) loop (i.e., before the outer iteration number 7). Please, show the entire content of the array as a sequence of integer values separated by a single space. No other symbols must be included in the response. This is an example of the response: 0 3 0 6 8 etc.

Risposta:

Domanda 2
Risposta non ancora data
Punteggio max.: 1,50
Contrassegna domanda
Modifica domanda

THIS EXERCISE IS ONLY FOR STUDENTS WITH 10 CREDITS IN THEIR CURRICULUM.

Given the following undirected and weighted graph find a minimum spanning tree using Prim algorithm. Start from vertex A.



Indicate the total weight of the final minimum spanning tree. Report one single integer value. No other symbols must be included in the response. This is an example of the response format: 13

Risposta:

Domanda 3
Risposta non ancora data
Punteggio max.: 1,50
Contrassegna domanda
Modifica domanda

The following sequence of pairs is given. Each relation i-j means that node i is adjacent to node j. Node names are integer values within the range from 0 to 9.

```
3-1 4-2 3-6 0-5 3-9 6-9 6-0
```

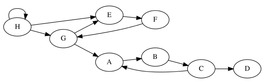
Apply an online connectivity algorithm based on quick-find. Please, show the content of the array at the end of the entire process as a sequence of integer values separated by a single space. No other symbols must be included in the response. This is an example of the response: 0 3 0 6 8 etc.

Risposta:

Domanda 4
Risposta non ancora data
Punteggio max.: 2,00
Contrassegna domanda
Modifica domanda

THIS EXERCISE IS ONLY FOR STUDENTS WITH 12 CREDITS IN THEIR CURRICULUM.

Given the following directed graph, represent the reverse graph and find all strongly connected components.



Compute all strongly connected components in the graph. Report them in alphabetic order and within each component indicates vertices in alphabetic order (i.e., ACZ and not CAZ or ZAC). Separate components with a single space. No other symbols must be included in the response. This is an example of the response format: ABC EF XY etc.

Risposta:

Domanda 5
Risposta non ancora data
Punteggio max.: 1,50
Contrassegna domanda
Modifica domanda

Given the following sequence of integers stored into an array, turn it into a heap, assuming to use an array as an underlying data structure. Assume that, in the end, the largest value is stored at the heap's root. Then, execute the first two steps of heapsort on the heap built at the previous step.

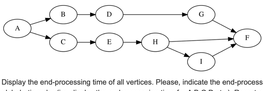
```
7 8 9 31 3 15 6 19 21 13
```

Report the final content of the entire array at the end of the above process. Please, show the entire content of the array as a sequence of integer values separated by a single space. No other symbols must be included in the response. This is an example of the response: 0 3 2 6 8 etc.

Risposta:

Domanda 6
Risposta non ancora data
Punteggio max.: 1,50
Contrassegna domanda
Modifica domanda

Visit the following graph in depth-first, starting at node A. Label nodes with discovery and end-processing times. Start with the discovery time set to 1 on A. When necessary, consider nodes and edges in alphabetic order.



Display the end-processing time of all vertices. Please, indicate the end-processing time of all vertices sorted in alphabetic order (i.e. display the end-processing time for A B C D etc.). Report a sequence of integer values separated by one single space. No other symbols must be included in the response. This is an example of the response: 15 13 2 16 8 etc.

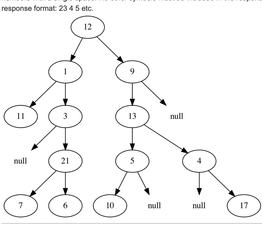
Risposta:

Domanda 7
Risposta non ancora data
Punteggio max.: 0,50
Contrassegna domanda
Modifica domanda

THIS EXERCISE IS ONLY FOR STUDENTS WITH 10 CREDITS IN THEIR CURRICULUM.

The following binary tree is given.

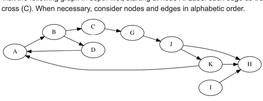
Visit it in post-order. Please, report the sequence of keys displayed by the visit on the same line and separate numbers with a single space. No other symbols must be included in the response. This is an example of the response format: 23 4 5 etc.



Risposta:

Domanda 8
Risposta non ancora data
Punteggio max.: 1,50
Contrassegna domanda
Modifica domanda

Visit the following graph in depth-first, starting at node A. Label each edge as tree (T), back (B), forward (F), and cross (C). When necessary, consider nodes and edges in alphabetic order.



Report the label of the edges BD, DA, and IH in this order. Please, indicate the edge type of these 3 arcs with a single letter, i.e., T, F, B, C, separated by single spaces. No other symbols must be included in the response. This is an example of the response: F T B

Risposta:

Domanda 9
Risposta non ancora data
Punteggio max.: 1,50
Contrassegna domanda
Modifica domanda

Consider a binary tree with 13-nodes. Its visits return the following sequences.

```
Pre-order:    M  Z  X  A  D  H  L  K  S  W  V  Y  P
In-order:     X  A  Z  D  H  L  M  S  K  Y  V  W  P
Post-order:   A  X  L  H  D  Z  S  Y  V  P  W  K  M
```

Report the sequence of keys stored on the leaves of the tree, moving on the tree from left to right. Please, report the list of keys on the same line, separated by a single space. No other symbols must be included in the response. This is an example of the response format: A X C Y etc.

Risposta:

Domanda 10
Risposta non ancora data
Punteggio max.: 1,50
Contrassegna domanda
Modifica domanda

Insert the following sequence of keys into an initially empty hash table. The hash table has a size equal to 19. Insertions occur character by character using open addressing with quadratic probing (with c1=1 and c2=1). Each character is identified by its index in the English alphabet (i.e., A=1, ..., Z=26). Equal letters are identified by a different subscript (i.e., A and A become A1 and A2).

```
A B B B A
```

Indicate in which elements are placed the last two letters of the sequence, i.e., the last B and the last A, in this order. Please, report your response as a sequence of integer values separated by one single space. No other symbols must be included in the response. This is an example of the response format: 3 4

Risposta:

Domanda 11
Risposta non ancora data
Punteggio max.: 1,00
Contrassegna domanda
Modifica domanda

Given an initially empty BST, perform the following sequence of operations on the BST **leaves**. Each positive value indicates a **leaf** insertion and each negative value a node extraction.

```
9  30  3  0  7  -1  16  17  14  21  13  29
```

Indicate the structure of the final BST reporting all nodes' key from top to bottom (from root to leaves) and from left to right (for each tree level). Please, report all nodes' key as a sequence of integer values separated by a single space. No other symbols must be included in the response. This is an example of the response format: 23 4 5 3 etc.

Risposta:

Domanda 12
Risposta non ancora data
Punteggio max.: 1,50
Contrassegna domanda
Modifica domanda

Given the following array of integer values, sort it in ascending order using the shell sort procedure. Use the Knuth's sequence h = 3h + 1, i.e., 1, 4, 13, etc.

```
13 9 18 11 -2 5 3 1 7 4 18 3 2 21 5
```

Display the content of the array just before the main (outer) loop iteration with h=1. Please, show the entire content of the array as a sequence of integer values separated by a single space. No other symbols must be included in the response. This is an example of the response: 0 3 0 6 8 etc.

Risposta:

Successivo